BG770A-GL&BG95xA-GL FILE Application Note

LPWA Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2021-02-04	Damir KLICKOVIC	Creation of the document
1.0	2022-05-05	Damir KLICKOVIC	First official release

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1 Introduction

Quectel BG770A-GL, BG950A-GL and BG951A-GL modules support AT commands for the operation of files on different physical storage mediums. This document is an application note to these commands.

The supported storage mediums are:

- **UFS**: User File Storage directory on the modem side; it is a special directory in the flash file system.
- **EUFS**: Extended User File Storage on the application side.

The file name indicates storage location. A file name beginning with the prefix “UFS:” or without any prefix indicates that the file is in the UFS, while the one beginning with the prefix “EUFS:” indicates that it is in the EUFS.

1.1. Using FILE AT Commands

The general procedure for uploading/downloading, opening/creating, reading and writing to a file in the storage:

Step 1: Upload a file to the storage with **AT+QFUPL**. If necessary, download the file with **AT+QFDWL** to verify its content, data integrity, etc.

Step 2: Open the file with **AT+QFOPEN**. When the file is opened, you can write to it or read from it any time and from any location until the file is closed with **AT+QFCLOSE**.
When opening a file with **AT+QFOPEN**, you can, for example, set the file into overwrite mode or read-only mode with **<mode>** parameter (see ***Chapter 2.3.6*** for detailed information). After the file is opened, a **<filehandle>** is assigned to it, so that various file operations can be carried out.

Step 3: After the file is opened, you can write to it with **AT+QFWRITE** or read the data content from the current file pointer position with **AT+QFREAD**.
- You can set the file pointer position with **AT+QFSEEK** or query the current position with **AT+QFPOSITION**.
- **AT+QFTUCAT** will truncate the file from the current position to the end of the file.

Step 4: Close the file with **AT+QFCLOSE**, after which the **<filehandle>** becomes invalid.

Commonly used commands to manage files in the storage:

1) **AT+QFLDS**: Get storage space information.
2) **AT+QFLST**: List the file information in the specified storage.
3) **AT+QFDEL**: Delete the file(s) in the specified storage.

1.2. Description of Data Mode

The COM port of the BG770A-GL, BG950A-GL and BG951A-GL modules has two working modes: AT command mode and data mode. In the AT command mode, the data inputted via the COM port are viewed as AT commands, whereas in the data mode, they are treated as data.

- **Enter Data Mode**

Execute **AT+QFUPL**, **AT+QFDWL**, **AT+QFREAD** or **AT+QFWRITE** to enter the data mode. If you input **+++** or pull the MAIN_DTR pin up to make the port exit the data mode, the execution of these commands will be interrupted before the response is returned. In such a case, the COM port cannot re-enter the data mode if you execute **ATO**.

- **Exit Data Mode**

Input **+++** or pull the MAIN_DTR pin up to make the COM port exit the data mode. To prevent the **+++** from being misinterpreted as data, the following sequence should be followed:

1) Do not input any character for at least 1 second before and after inputting **+++**.
2) Input **+++** within 1 second, and wait until **OK** is returned. Once **OK** is returned, the COM port exits the data mode and switches to AT command mode.

If you are exiting the data mode by pulling the MAIN_DTR pin up, make sure to set **AT&D1** first.

2 Description of FILE AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	**AT+<cmd>=?**	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	**AT+<cmd>?**	Check the current parameter value of the corresponding command.
Write Command	**AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]**	Set user-definable parameter value.

Execution Command	**AT+<cmd>**	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. AT Command Description

2.3.1. AT+QFLDS Get Space Information of Storage Medium

This command gets the space information of the specified storage medium.

AT+QFLDS Get Space Information of Storage Medium	
Test Command **AT+QFLDS=?**	Response **OK**
Write Command **AT+QFLDS=<storage>**	Response **+QFLDS: <freesize>,<total_size>** **OK** If there is an error related to the ME functionality: **+CME ERROR: <err>**
Execution Command **AT+QFLDS**	Response **+QFLDS: <UFS_file_size>,<UFS_file_number>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<storage>	String type. Storage medium type. "UFS" UFS on modem side "EUFS" Extended UFS on application side
<freesize>	Integer type. Free space size of **<storage>**. Unit: byte.
<total_size>	Integer type. Total size of **<storage>**. Unit: byte.
<UFS_file_size>	Integer type. Size of all files in UFS. Unit: byte
<UFS_file_number>	Integer type. Number of files in UFS.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

Example

```
AT+QFLDS="UFS"                //Query the space information of UFS on modem side.
+QFLDS: 1249984,1562624

OK
AT+QFLST="*"
+QFLST: "Test1.txt",10
+QFLST: "Test2.txt",16
+QFLST: "Test3.txt",150
+QFLST: "Test4.txt",2056
+QFLST: "Test5.txt",98

OK
AT+QFLDS                      //Query the size and number of all files in UFS.
+QFLDS: 2330,5

OK
```

2.3.2. AT+QFLST List File Information on Storage Medium

This command lists the information of a single file or all files on a specified storage medium.

AT+QFLST List File Information on Storage Medium	
Test Command **AT+QFLST=?**	Response **OK**
Write Command **AT+QFLST=<filename>**	Response **+QFLST: <file>,<file_size>** **[+QFLST: <file>,<file_size>** **[…]]** **OK**

	If there is any error: **+CME ERROR: <err>**
Execution Command **AT+QFLST**	Response Return the information of UFS files: **+QFLST: <file>,<file_size>** **[+QFLST: <file>,<file_size>** **[…]]** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filename>	String type. File to be listed.	
	"*"	All files in UFS
	"UFS:*"	All files in UFS
	"EUFS:/ufs/*"	All files in the *ufs* directory of EUFS
	"EUFS:/datatx/*"	All files in the *datatx* directory of EUFS
	"EUFS:*"	All files in the *ufs* directory of EUFS
	"**<file>**"	A specified file **<file>** in UFS
	"UFS:**<file>**"	A specified file **<file>** in UFS
	"EUFS:**<file>**"	A specified file **<file>** in *ufs* directory of EUFS
	"EUFS:/ufs/**<file>**"	A specified file **<file>** in *ufs* directory of EUFS
	"EUFS:/datatx/**<file>**"	A specified file **<file>** in *datatx* directory of EUFS
<file>	String type. Exact name of file. Maximum length: 28 bytes.	
<file_size>	Integer type. File size. Unit: byte.	
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.	

Example

```
AT+QFLST="*"                          //List all files in UFS.
+QFLST: "F_M12-1.bmp",562554
+QFLST: "F_M12-10.bmp",562554
+QFLST: "F_M12-11.bmp",562554

OK
AT+QFLST="Test1.txt"                  //List a specified file "Test1.txt" in UFS.
```

```
+QFLST: "Test1.txt",2

OK
AT+QFLST="EUFS:/ufs/*"                  //List all files in ufs directory of EUFS .
+QFLST: "EUFS:test.txt",10
+QFLST: "EUFS:test2.txt",20

OK
AT+QFLST="EUFS:test.txt"                //List a specified file "test.txt" in ufs directory of EUFS.
+QFLST: "EUFS:test.txt",10

OK
```

2.3.3. AT+QFDEL Delete File(s) on Storage Medium

This command deletes a specified file or all files from a storage medium.

AT+QFDEL Delete File(s) on Storage Medium	
Test Command **AT+QFDEL=?**	Response **+QFDEL: <filename>** **OK**
Write Command **AT+QFDEL=<filename>**	Response **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filename>	String type. Name of the file to be deleted. Maximum length: 28 bytes.	
	"*"	All files in UFS
	"UFS:*"	All files in UFS
	"EUFS:/ufs/*"	Delete all files in the *ufs* directory of EUFS (*ufs* directory remains)
	"EUFS:/datatx/*"	Delete all files in the *datatx* directory of EUFS (*datatx* directory remains)
	"EUFS:*"	Delete all files in the *ufs* directory of EUFS (*ufs* directory remains)
	"UFS:**<file>**"	A specified file **<file>** in UFS
	"EUFS:**<file>**"	Delete a specified file **<file>** in *ufs* directory of EUFS

	"EUFS:/ufs/**<file>**"	Delete a specified file **<file>** in *ufs* directory of EUFS
	"EUFS:/datatx/**<file>**"	Delete a specified file **<file>** in *datatx* directory of EUFS
<file>	String type. Exact name of file.	
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.	

Example

```
AT+QFDEL="*"
OK

AT+QFDEL="Test1.txt"
OK
```

2.3.4. AT+QFUPL Upload File to Storage Medium

This command uploads a file to a storage medium. If any file on the storage medium has the same name as the uploaded file, an error will be reported.

After the Write Command is executed and **CONNECT** is returned, the module will switch to the data mode. When the uploaded data reach **<file_size>**, or if no other data are inputted when **<timeout>** is reached, the module exits the data mode automatically. During data transmission, you can send **+++** or pull the MAIN_DTR pin high to make the module exit the data mode. For more information, see ***Chapter 1.2.***

AT+QFUPL Upload File to Storage Medium	
Test Command **AT+QFUPL=?**	Response **+QFUPL: <filename>[,(1-<freesize>)[,(**range of supported **<timeout>**s**)[,(**list of supported **<ackmode>**s**)]]]** **OK**
Write Command **AT+QFUPL=<filename>[,<file_size>[,<timeout>[,<ackmode>]]]**	Response **CONNECT** The TA switches to the data mode (transparent transmission mode), so the binary data of the file can be inputted. When the total size of the inputted data reaches **<file_size>** or TA receives **+++**, TA will return to the command mode and report the following code: **+QFUPL: <upload_size>,<checksum>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately;

	The configurations will not be saved.

Parameter

<freesize>	Integer type. Free space size of **<storage>**. See **AT+QFLDS** for more information about **<storage>**.
<filename>	String type. Name of the file to be uploaded. Maximum length: 28 bytes. "**<file>**" Upload the file to UFS "UFS:**<file>**" Upload the file to UFS "EUFS:**<file>**" Upload the file to *ufs* directory of EUFS "EUFS:/ufs/**<file>**" Upload the file to *ufs* directory of EUFS "EUFS:/datatx/**<file>**" Upload the file to *datatx* directory of EUFS
<file>	String type. The exact name of the file.
<file_size>	Integer type. File size expected to be uploaded. Default value: 10240. Unit: byte. Maximum length is not greater than **<freesize>**.
<upload_size>	Integer type. Actual size of uploaded data. Unit: byte.
<timeout>	Integer type. Waiting time for inputting data to UART and USB. Range: 1–65535. Default value: 5. Unit: s.
<ackmode>	Integer type. Determines whether to use the ACK mode. <u>0</u> Turn ACK mode off. 1 Turn ACK mode on.
<checksum>	Integer type. Checksum of uploaded data.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

1. It is strongly recommended to use DOS 8.3 file name format for **<filename>**.
2. **<checksum>** is a 16-bit checksum based on bitwise XOR.
   When the number of characters is odd, the last character is set as the high 8 bit, and the low 8 bit as 0, and then an XOR operator is used to calculate the checksum. Inputting **+++** will make the TA end the data transmission and switch to the command mode. However, the previously uploaded data will be preserved in the file.
3. When executing the command, the data must be entered after **CONNECT** is returned.
4. The ACK mode is a safeguard against data loss when uploading a large file, if hardware flow control does not work. The ACK mode works as follows:
   1) Run **AT+QFUPL=<filename>,<file_size>,<timeout>,1** to enable the ACK mode.
   2) The module outputs **CONNECT**.
   3) The MCU sends 1 K bytes of data, to which the modules respond with an **A**.
   4) The MCU receives the **A** and then sends the next 1 K bytes of data.
   5) Steps 3 and 4 are repeated until the transfer is completed.
   For an example of ACK mode use, see ***Chapter 3.1.1.2***

5. Due to the physical limitations of the platform, i.e., write speed to the NOR flash memory, the upload feature has 2 operational modes to guarantee that the entire file will be written without data loss. Two modes are: 1) without activated HW flow control 2) with activated HW flow control.
   1) When the HW flow control is not activated, the maximum file size is limited to 200000 bytes. Attempting to send files larger than that will result in an error. Writing time can significantly vary depending on file size, content, and area of the flash where the file is being written.
   2) When HW flow control is activated, by executing **AT+IFC=2,2**, there is no file size limitation other than the available disk space. To ensure safe file transfer in this mode, on the client side, RTS and CTS must be switched on before sending the file. Transmission time can be longer than in the mode without activated HW flow control.

2.3.5. AT+QFDWL Download File from Storage Medium

This command downloads a specified file from a storage medium.

AT+QFDWL Download File from Storage Medium	
Test Command **AT+QFDWL=?**	Response **+QFDWL: <filename>** **OK**
Write Command **AT+QFDWL=<filename>**	Response **CONNECT** The TA switches to the data mode, so the binary data of the file can be outputted. When the content of the file is read or the TA receives **+++**, the TA will return to the command mode and report the following code:: **+QFDWL: <download_size>,<checksum>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filename>	String type. Name of the file to be downloaded. Maximum length: 28 bytes.	
	"**<file>**"	Download the file from UFS
	"UFS:**<file>**"	Download the file from UFS
	"EUFS:**<file>**	Download the file from the *ufs* directory of EUFS
	"EUFS:/ufs/**<file>**"	Download the file from the *ufs* directory of EUFS
	"EUFS:/datatx/**<file>**"	Download the file from the *datatx* directory of EUFS

<file>	String type. The exact name of the file.
<download_size>	Integer type. Size of downloaded data. Unit: byte.
<checksum>	Integer type. Checksum of downloaded data.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

1. Inputting **+++** causes the TA to end the data transmission and switch to the command mode.
2. **<checksum>** is a 16-bit checksum based on bitwise XOR.

2.3.6. AT+QFOPEN Open a File

This command opens a file and gets the file handle to be used in commands such as **AT+QFREAD**, **AT+QFWRITE**, **AT+QFSEEK**, **AT+QFPOSITION**, **AT+QFTUCAT** and **AT+QFCLOSE**.

AT+QFOPEN Open a File	
Test Command **AT+QFOPEN=?**	Response **+QFOPEN: <filename>[,(**range of supported <**mode**>s**)]** **OK**
Read Command **AT+QFOPEN?**	Response **+QFOPEN: <filename>,<filehandle>,<mode>** **[+QFOPEN: <filename>,<filehandle>,<mode>** **[…]]** **OK**
Write Command **AT+QFOPEN=<filename>[,<mode>]**	Response **+QFOPEN: <filehandle>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filename>	String type. Name of the file to be opened. Maximum length: 28 bytes.	
	"**<file>**"	Open the file in UFS
	"UFS:**<file>**"	Open the file in UFS
	"EUFS:**<file>**"	Open the file of the *ufs* directory of EUFS.

	"EUFS:/ufs/**<file>**"	Open the file of the *ufs* directory of EUFS
	"EUFS:/datatx/**<file>**"	Open the file of the *datatx* directory of EUFS
<file>	String type. The exact name of the file.	
<filehandle>	Integer type. Handle of the file to be used.	
<mode>	Integer type. File opening mode. 0 If the file does not exist, it is created. If the file exists, it is opened directly. In both cases, the file can be read and written to. 1 If the file does not exist, it is created. If the file exists, it is overwritten. In both cases, the file can be read and written to. 2 If the file exists, it is opened directly as a read-only file. Otherwise, an error is returned. 3 If the file does not exist, it is created. If the file exists, data are appended at the end of the file. In both cases, the file can be read and written to.	
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.	

NOTE

<filehandle> starts form 0 in UFS, whereas it starts from 20000 in EUFS.

2.3.7. AT+QFREAD Read a File

This command reads the data of a file specified by the file handle. The data start from the current position of the file pointer that belongs to the file handle.

AT+QFREAD Read a File	
Test Command **AT+QFREAD=?**	Response **+QFREAD: <filehandle>[<data_length>]** **OK**
Write Command **AT+QFREAD=<filehandle>[<data_length>]**	Response **CONNECT <read_length>** The TA switches to the data mode. When the total size of the data reaches **<data_length>** (unit: byte), the TA will go back to the command mode, display the result and report the following code: **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file to be read.
<data_length>	Integer type. Length of the file to be read. Default value: file length. Unit: byte.
<read_length>	Integer type. Actual read length. Unit: byte.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

2.3.8. AT+QFWRITE Write to a File

This command writes data into a file. The data start from the current position of the file pointer that belongs to the file handle.

AT+QFWRITE Write to a File	
Test Command **AT+QFWRITE=?**	Response **+QFWRITE: <filehandle>[<data_length>[,<timeout>]]** **OK**
Write Command **AT+QFWRITE=<filehandle>[<data_length>[,<timeout>]]**	Response **CONNECT** The TA switches to the data mode. When the total size of the written data reaches **<data_length>** (unit: byte) or the time reaches **<timeout>**, the TA will go back to the command mode and report the following code: **+QFWRITE: <written_length>,<total_length>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file to be written to.
<data_length>	Integer type. Length of the file to be written to. Maximum value of this parameter is determined by **<freesize>** of **AT+QFUPL**. Default value: 10240. Unit: byte.
<timeout>	Integer type. Waiting time for inputting data to USB/UART. Range: 1–65535. Default value: 5. Unit: s.
<written_length>	Integer type. Actual written length. Unit: byte.
<total_length>	Integer type. Total file length. Unit: byte.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

1. Due to the physical limitations of the platform, i.e., write speed to the NOR flash memory, write feature has 2 operational modes to guarantee that the entire file will be written without data loss. Two modes are: 1) without activated HW flow control 2) with activated HW flow control.
   1) If the HW flow control is not activated, the maximum file size is limited to 200000 bytes. Attempting to send files larger than that will result in an error. Writing time can vary significantly depending on file size, content, and area of the flash where the file is being written.
   2) When HW flow is activated, by executing **AT+IFC=2,2**, there is no file size limitation other than the available disk space. To ensure safe file transfer in this mode, on the client side, the RTS and CTS must be switched on before sending the file. Transmission time can be longer than in the mode without activated HW flow control.

2.3.9. AT+QFSEEK Set File Pointer to a Position

This command sets a file pointer to a specified position. This will decide the starting position of commands such as **AT+QFREAD**, **AT+QFWRITE**, **AT+QFPOSITION** and **AT+QFTUCAT**.

AT+QFSEEK Set File Pointer to a Position	
Test Command **AT+QFSEEK=?**	Response **+QFSEEK: <filehandle>,<offset>[,<position>]** **OK**
Write Command **AT+QFSEEK=<filehandle>,<offset>[,<position>]**	Response **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file for which a pointer is set.
<offset>	Integer type. Number of bytes of the file pointer movement.
<position>	Integer type. Pointer movement mode. 0 Move forward from the beginning of the file. 1 Move forward from the current position of the pointer. 2 Move backward from the end of the file.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

1. If **<position>**=0/2 and **<offset>** exceed the file size, the command returns **ERROR**.
2. If **<position>**=1 and the total size of **<offset>** and the current position of the pointer exceeds the file size, the command returns **ERROR**.

2.3.10. AT+QFPOSITION Get Offset of File Pointer

This command gets the offset of a file pointer from the beginning of the file.

AT+QFPOSITION Get Offset of File Pointer	
Test Command **AT+QFPOSITION=?**	Response **+QFPOSITION: <filehandle>** **OK**
Write Command **AT+QFPOSITION=<filehandle>**	Response **+QFPOSITION: <offset>** **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file for which the offset of file pointer is to be gotten.
<offset>	Integer type. Offset from the beginning of the file.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

2.3.11. AT+QFTUCAT Truncate File from File Pointer

This command truncates the file from the current pointer position to the end of the file.

AT+QFTUCAT Truncate File from File Pointer	
Test Command **AT+QFTUCAT=?**	Response **+QFTUCAT: <filehandle>** **OK**
Write Command **AT+QFTUCAT=<filehandle>**	Response **OK**

	If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file to be truncated.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

2.3.12. AT+QFCLOSE Close a File

This command closes a file and ends all file operations. After that, the file handle is released and should not be used again, unless the file is re-opened with **AT+QFOPEN**.

AT+QFCLOSE Close a File	
Test Command **AT+QFCLOSE=?**	Response **+QFCLOSE: <filehandle>** **OK**
Write Command **AT+QFCLOSE=<filehandle>**	Response **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filehandle>	Integer type. Handle of the file to be closed.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

2.3.13. AT+QFCRC Calculate CRC of a Specified UFS File

This command calculates the CRC16, CRC16_CCITT and CRC32 checksums of a specified UFS file.

AT+QFCRC Calculate CRC of a Specified UFS File	
Test Command **AT+QFCRC=?**	Response **+QFCRC: <filename>**

	OK
Write Command **AT+QFCRC=<filename>**	Response When the specified file exists, the command returns 32-bit CRC, 16-bit CRC values and 16-bit CRC CCITT: **+QFCRC: <crc32>,<crc16>,<crc16_ccitt>** **OK** When the file exists but has no bytes: **+QFCRC: 0,0,0** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<filename>	String type. Input format: "UFS:filename". "UFS:" can be omitted.
<crc32>	32-bit CRC value.
<crc16>	16-bit CRC value.
<crc16_ccitt>	16-bit CRC CCITT value.
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

This command applies to UFS files only.

2.3.14. AT+QFCPY Make a Copy of a Specified File

This command copies the whole content of a source file to a destination file.

- If the destination file already exists, the command returns an **ERROR** unless **<enable>**=1.
- If the destination file has the same name as the source file, the command returns an **ERROR**.
- If the remaining space of the file system is smaller than the file size to be copied, the command also returns an **ERROR**.

AT+QFCPY Make a Copy of a Specified File	
Test Command **AT+QFCPY=?**	Response **+QFCPY: <src_file>,<dest_file>[,<enable>]**

	OK
Write Command **AT+QFCPY=<src_file>,<dest_file>[,<enable>]**	Response **OK** If there is any error: **+CME ERROR: <err>**
Characteristics	The command takes effect immediately; The configurations will not be saved.

Parameter

<src_file>	String type. Source file name. Input format: "UFS:filename". "UFS:" can be omitted.
<dest_file>	String type. Destination file name. Input format: “UFS:filename”. “UFS:” can be omitted.
<enable>	Integer type. Determines whether to overwrite the existing file that has the same name as the destination file. <u>0</u> Disable 1 Enable
<err>	Integer type. Error code. See ***Chapter 4*** for possible **<err>** values.

NOTE

This command applies to UFS files only.

3 Examples

3.1. Upload and Download Files

3.1.1. Upload a File

3.1.1.1. Non-ACK Mode

AT+QFUPL="test1.txt",10	//Upload the text file "test1.txt" to UFS.
CONNECT	
<Input file bin data>	
+QFUPL: 10,3938	
OK	
AT+QFUPL="EUFS:test.txt",4	//Upload the text file "test.txt" to *ufs* directory of EUFS.
CONNECT	
<Input file bin data>	
+QFUPL: 4,6a05s	
OK	

3.1.1.2. ACK Mode

The ACK mode can make data transmission more reliable. When transmitting a large file without hardware flow control, the ACK mode is recommended to prevent the data loss. For more information about the ACK mode, see **AT+QFUPL**.

AT+QFUPL="test.txt",3000,5,1	//Upload the text file "test.txt" to UFS.
CONNECT	
<input file bin data of 1024bytes>	
A	//After receiving 1024 bytes of data, the module returns **A**. Then the next 1024 bytes of data can be inputted.
<input file bin data of 1024bytes>	
A	
<input the rest file bin data>	

```
+QFUPL: 3000,B34A

OK
```

3.1.2. Download a File

```
AT+QFDWL="test.txt"          //Download the text file "test.txt" from UFS.
CONNECT
<Output Data>
+QFDWL: 10,613e              //Size and checksum value of the downloaded data are
                             returned.

OK
```

3.2. Write to and Read Files

3.2.1. Write to and Read a UFS File

```
AT+QFOPEN="test",0           //Open the file to get the file handle.
+QFOPEN: 0

OK
AT+QFWRITE=0,10              //Write 10 bytes to the file.
CONNECT
<Write Data>
+QFWRITE: 10,10              //The actual written bytes and the size of the file are returned.

OK
AT+QFSEEK=0,0,0              //Set the file pointer to the beginning of the file.
OK
AT+QFREAD=0,10               //Read 10 bytes from the file.
CONNECT 10
<Read Data>

OK
AT+QFCLOSE=0                 //Close the file.
OK
```

3.2.2. Write to and Read an EUFS File

```
AT+QFLDS="EUFS"              //Query the space information of EUFS.
+QFLDS: 1388544,2435072
```

```
OK
AT+QFOPEN="EUFS:test",0                  //Open the file to get the file handle.
+QFOPEN: 20000

OK
AT+QFWRITE=20000,10                      //Write 10 bytes to the file.
CONNECT
<Write Data>
+QFWRITE: 10,10                          //The actual written bytes and the size of the file are returned.

OK
AT+QFSEEK=20000,0,0                      //Set the file pointer to the beginning of the file.
OK
AT+QFREAD=20000,10                       //Read 10 bytes from the file.
CONNECT 10
<Read Data>

OK
AT+QFCLOSE=20000                         //Close the file.
OK
```

4 Summary of Error Codes

The error code **<err>** indicates an error related to mobile equipment. The detailed information about **<err>** is presented in the following table, which includes only the error codes related to the file operation of the module.

Table 2: Summary of Error Codes

<err>	Meaning
400	Invalid input value
401	Larger than the size of the file
402	Zero-byte read
403	Drive full
405	File not found
406	Invalid file name
407	File already exists
409	Failed to write to the file
410	Failed to open the file
411	Failed to read the file
413	Reached the max. number of files allowed to be opened
414	The read-only file
416	Invalid file descriptor
417	Failed to list the file
418	Failed to delete the file
419	Failed to get disk info

420	No space
421	Time-out
423	File too large
425	Invalid parameter
426	File already opened
427	Failed to allocate memory. If the file size is >200KB, IFC: 2,2 must be used
428	Maximum allowed number of files on the drive exceeded

5 Appendix Reference

Table 3: Terms and Abbreviations

Abbreviation	Description
ACK	Acknowledgement
COM	Communication Port
CRC	Cyclic Redundancy Check
DOS	Disk Operating System
EUFS	Extended User File Storage
MCU	Microprogrammed Control Unit
ME	Mobile Equipment
TA	Terminal Adapter
UART	Universal Asynchronous Receiver/Transmitter
UFS	User File Storage
USB	Universal Serial Bus
XOR	Exclusive OR